

定积分应用

考试大纲

数学一、二

掌握用定积分表达和计算一些几何量与物理量

函数的平均值

平面图形的面积、平面曲线的弧长、旋转体体积及侧面积、平行截面面积为已知的立体体积

功、引力、压力、质心、形心等

数学三

会利用定积分计算平面图形的面积、旋转体体积和函数的平均值

会利用定积分求解简单的经济应用问题

几何应用

微元法

取任意一点x 波动一下x+dx

取一小块近似研究

积分微元加

平均值

平均值=区间积分÷区间长度

$$\text{平均值} = \frac{1}{b-a} \int_a^b f(x) dx$$

面积

直角坐标

$$s = \int_a^b |f(x)| dx$$

注意：单侧函数可以转化为双侧函数
(例如出现绝对值可以转化为指数)
把x写成y的函数 对y积分

参数方程

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(x(t)) x'(t) dt$$

极坐标

$$s = \int_a^b \sqrt{r^2 + (r')^2} dt$$

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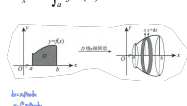
参数方程

$$\int_a^b f(x) dx = \int_{\alpha}^{\beta} f(x(t)) x'(t) dt$$

旋转体体积

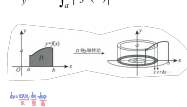
绕x轴旋转

$$V_x = \pi \int_a^b f^2(x) dx$$



绕y轴旋转

$$V_y = 2\pi \int_a^b |x| f(x) dx$$



绕xy轴平行线转 (平移)

微元法

注意：单侧函数可以转化为双侧函数

任意一点x，波动y+dy，取一小块近

似，反映x，算出微元的体积，对y积分

二重积分

绕斜线转

二重积分

弧长

直角坐标

公式：

(1) y(x)

$$s = \int_a^b \sqrt{1 + (y')^2} dx$$

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▼ 定积分应用

- 考试大纲

▼ 几何应用

- 微元法
- 平均值

▼ 面积

- 直角坐标
- 参数方程
- 极坐标

▼ 旋转体体积

- 绕x轴旋转
- 绕y轴旋转
- 绕x/y轴平行线转（平移）
- 绕斜线转

▼ 弧长

- 直角坐标
- 参数方程
- 极坐标

▼ 侧面积

- 直角
- 参数
- 极坐标

▼ 物理应用（数一数二的）

▼ 受力

- 万有引力
- 液体压力

- 做功

▼ 形心质心

- 面
- 线